

Project Name: **P07 HDC Project**

Unit Tag: **CH-8**

Qty.: **1**

Model: **YKGEELP85ERHS**

Full Load - Design

PIN

YKGEELP85E	RHSRVGNE3V	VB0569000X	XXCCHRX1CL	FXXW00XX1C	LFXXW00XXX	XXX33XGGNX	XXXXXSXYX	XBEXTXXJXX	XXXXXXHXXA	XXIWS658
....5...10	5...20	5...30	5...40	5...50	5...60	5...70	5...80	5...90	5...100	5...110

Unit

Model No.	YKGEELP85ERHS
Number of Compressors	1
Compressor Type	Centrifugal
Number of Compressor Circuits	1
Refrigerant	R-515B
Gear	RV
Variable Orifice	3

Performance Data

Specified Net Capacity [kW]	2000
Rated Net Capacity [kW]	2000
Full Load Efficiency [kW/kW]	4.276
Heat Rejection Capacity [kW]	2431
A-Weighted Sound Pressure Level [dB(A)]	85.0

Electrical Data

Power Supply [V/ph/Hz]	380/3/50.0
Total Input Power [kW]	467.7
Job FLA [A]	718.0
LRA [A]	5600.0
Min. Circuit Ampacity [A]	898.0
Max. Circuit Breaker Amps [A]	1600.0
Unit Short Circuit Withstand (STD) [kA]	100
Motor Power Supply [V/ph/Hz]	380/3/50.0
Motor FLA [A]	798.0
Max. Motor Load [kW]	473.0
Motor Input Power [kW]	453.2
Motor Output Power [kW]	425.1
Oil Pump Voltage [V]	380/3/50.0



Performance Impacting Options

Starter Type	Unit Mounted Low Voltage Water Cooled VSD w/ Filter
Starter Model	VSD658KFT-50
VSD Cooling Type	Evaporator Watercooled Starter
Condenser Gas Inlet Type	Condenser Gas Inlet Baffle
Oil Cooler	Refrigerant Cooled Oil Cooler
Isolation Valves	Yes
Optisound Control	Yes

Weight & Dimensional Data

Shipping Weight [kg]	11013
Operating Weight [kg]	12400
Refrigerant Charge [kg]	457

Project Name: P07 HDC Project

Unit Guid: acff4d24-6e52-4615-8372-6a0eb10a5286

Rating Engine Version: v1_236.idd

Unit Name: CH-8

Version: SN25.12a

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Performance Report

Performance Specification

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Project Name: **P07 HDC Project**Unit Tag: **CH-8**Qty.: **1**Model: **YKGEELP85ERHS**

Heat Exchanger Performance

Evaporator		Condenser	
Fluid Type*	Water	Fluid Type*	Water
Tube MTI No.	361	Tube MTI No.	431 / 431
Passes*	2	Passes*	2
Fouling Factor* [m2 K/kW]	0.0176	Fouling Factor* [m2 K/kW]	0.0440
Entering Fluid Temperature* [°C]	25.00	Entering Fluid Temperature* [°C]	52.00
Leaving Fluid Temperature* [°C]	19.00	Leaving Fluid Temperature* [°C]	58.00
Flow Rate [L/s]	79.88	Flow Rate [L/s]	98.16
Pressure Drop [kPa]	30.5	Pressure Drop [kPa]	25.7
Minimum Flow Rate [L/s]	47.62	Minimum Flow Rate [L/s]	65.43
Maximum Flow Rate [L/s]	181.2	Maximum Flow Rate [L/s]	235.8

* Designates user specified input

Outside the scope of AHRI Water-Cooled Water-Chilling and Heat Pump Water-Heating Packages Certification Program or not optionally certified, but is rated in accordance with AHRI Standard 550/590 (I-P) and AHRI Standard 551/591 (SI). Auxiliary components included in total KW - oil pump & heater, chiller controls.

Part Load Performance (Based on Minimum Condenser Water Temperature)

CEFT [°C]	Percent Load									
	100.0	90.0	80.0	70.0	60.0	50.0	40.0	30.0	20.0	15.0
52.00	4.276	4.284	4.229	4.159	4.132	4.006	3.816	3.531	3.100	2.784
51.67	4.327	4.336	4.280	4.210	4.184	4.056	3.862	3.607	3.135	2.815
48.89	4.777	4.789	4.744	4.683	4.662	4.528	4.310	3.972	3.466	3.095
46.11	5.282	5.319	5.345	5.286	5.253	5.079	4.805	4.410	3.839	3.425
43.33	5.834	5.937	5.987	5.993	5.845	5.761	5.468	5.006	4.319	3.832
40.56	6.492	6.663	6.799	6.885	6.754	6.544	6.278	5.706	4.872	4.305
37.78	7.256	7.478	7.679	7.838	7.859	7.584	7.231	6.536	5.524	4.844
35.00	8.099	8.415	8.706	8.963	9.045	8.814	8.247	7.505	6.266	5.448
32.22	9.138	9.540	9.923	10.28	10.45	10.30	9.583	8.662	7.162	6.172
29.44	10.36	10.86	11.40	11.90	12.18	12.00	11.30	9.929	8.163	6.937
26.67	11.82	12.56	13.33	13.95	14.20	13.84	12.87	11.20	8.914	7.500
23.89	13.64	14.68	15.66	15.83	15.84	15.31	14.13	12.19	9.401	7.924

* Values are in kW/kW

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Sound Pressure Levels (Standard)

CEFT [°C]	Percent Load									
	100.0	90.0	80.0	70.0	60.0	50.0	40.0	30.0	20.0	15.0
52.00	85.0	84.0	83.0	81.5	80.5	79.5	79.0	79.0	79.5	79.5
51.67	85.0	84.0	83.0	81.5	80.5	79.5	79.0	79.0	79.5	79.5
48.89	84.5	83.0	82.0	81.0	79.5	78.5	78.0	78.0	78.0	78.5
46.11	84.0	82.5	81.0	80.0	79.0	78.0	78.0	78.0	78.0	78.0
43.33	83.5	82.0	81.0	80.0	79.0	78.0	78.0	78.0	78.0	78.0
40.56	83.0	82.0	81.0	80.0	79.0	78.0	78.0	78.0	78.0	78.0
37.78	83.0	82.0	81.0	80.0	79.0	78.0	78.0	78.0	78.0	78.0
35.00	83.0	82.0	81.0	80.0	79.0	78.0	78.0	78.0	78.0	78.0
32.22	83.0	82.0	81.0	80.0	79.0	78.0	78.0	78.0	78.0	78.0
29.44	83.0	82.0	81.0	80.0	79.0	78.0	78.0	78.0	78.0	78.0
26.67	83.0	82.0	81.0	80.0	79.0	78.0	78.0	78.0	78.0	78.0
23.89	83.0	82.0	81.0	80.0	79.0	78.0	78.0	78.0	78.0	78.0

The octave and A-weighted sound pressure levels are the levels expected to be obtained if measurements are performed in accordance with AHRI Standard 575, Method of Measuring Machinery Sound Within Equipment Rooms. Tolerances: The sound levels of identical unit selections can vary due to manufacturing tolerances and test repeatability. Variations of +/- 3dBA on the A-weighted levels and +/- 5dB on the octave band levels are possible.

Unit Configuration Details

	Evaporator	Condenser
Water Box Type	Compact	Compact
Waterside Design Working Pressure [psig]	150	150
Entering Water Nozzle @ Location	L	L
Leaving Water Nozzle @ Location	L	L

Weight Breakdown Details

Shipping Weight [kg]	11013	Operating Weight [kg]	12400	Refrigerant (R-515B) Weight [kg]	457
Driveline Weight [kg]	3551				

Warnings:

DesignCooling: CEFT OF 65.00 F(18.33 C) FOR PLV CALCULATION IS LESS THAN 72.28 F(22.38 C) MINIMUM REQUIRED BY EQUIPMENT FOR THIS APPLICATION.

DesignCooling: PLV CALCULATION YIELDS NO IPLV OR NPLV VALUE.

DesignCooling: H OIL IS REQUIRED.

Notes:

Compliant with the requirements of the LEED Energy and Atmosphere Enhanced Refrigerant Management Credit (EAc4).

Materials and construction per mechanical specifications - Form 160.76-EG1.

Compliant with IECC - 2012,2015,2018,2021.

The product image shown is for illustrative purposes only and is not representative of selected options.

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